Hyperspectral Image Super-Resolution via Dual-Domain Network Based on Hybrid Convolution

Tingting Liu, Yuan Liu, Chuncheng Zhang, Liyin Yuan, Xiubao Sui, and Qian Chen

***Abstract*— Hyperspectral images (HSIs) with high spatial resolution are challenging to obtain directly due to sensor limitations. Deep learning is able to provide an end-to-end reconstruction solution from low to high spatial resolution. Nevertheless, existing deep learning-based methods have two main drawbacks. First, deep networks with self-attention mechanisms often require a trade-off between internal resolution, model performance, and complexity, leading to the loss of fine-grained, high-resolution (HR) features. Second, there are visual discrepancies between the reconstructed HSI and the ground truth because they focus on spatial-spectral domain learning. In this article, a novel super-resolution (SR) algorithm for HSIs, named SRDNet, is proposed by using a dual-domain network with hybrid convolution and progressive upsampling to exploit both spatial-spectral and frequency information of the hyperspectral data. In this approach, we design a self-attentive pyramid structure (HSL) to capture interspectral self-similarity in the spatial domain, thereby increasing the receptive range of attention and improving the feature representation of the network. Additionally, we introduce a hyperspectral frequency loss (HFL) with dynamic weighting to optimize the model in the frequency domain and improve the perceptual quality of the HSI. Experimental results on three benchmark datasets show that SRDNet effectively improves the texture information of the HSI and outperforms state-of-the-art methods. The code is available at https://github.com/LTTdouble/SRDNet.**

***Index Terms*— Dual-domain network, frequency loss, hyperspectral image, self-attention mechanism, super-resolution (SR).**

I. Introduction

HYPERSPECTRAL image (HSI) captures surface features of the Earth across numerous contiguous spectral bands. It is applied to environmental monitoring, military reconnaissance, precision agriculture, and more [1], [2], [3]. In recent years, it has emerged as a research hotspot in remote sensing. However, a common challenge in hyperspectral imagers is balancing spatial and spectral resolution. To achieve higher spectral resolution, the spatial resolution is often compromised, resulting in low-quality HSI [4], [5]. The limited spatial resolution causes spectral mixing of end members, impacting HSI detection performance.

Image super-resolution (SR) involves generating high-resolution (HR) images from low-resolution (LR) counterparts, effectively bypassing the limitations of imaging systems [6], [7], [8]. HSI SR is mainly categorized into fusion-based and single HSI SR methods [9]. Fusion-based methods have been proven effective in some works [10], [11], [12], [13]. It aims to find and extract missing high-frequency details from HR panoramic (PAN) or multispectral images (MSIs). Nevertheless, it is challenging to obtain these auxiliary registration images in real scenes. In this context, a single HSI SR has been gaining attention in reality. To tackle the issue of limited information in a single LR HSI, this approach utilizes image priors from an extensive external dataset to effectively constrain the solution space for the ill-posed SR problem.

Lately, deep learning techniques have rapidly advanced and have led to remarkable breakthroughs in RGB image SR, particularly with the successful utilization of convolutional neural networks (CNNs) in image processing. Through deep learning models, complex mapping relationships can be learned from large amounts of training data to predict HR images. But compared to RGB image SR, HSI SR is more demanding [14]. First, HSIs comprise hundreds of spectral bands, complicating the consideration of spectral correlation when enhancing resolution. Second, HSIs encompass vast geographic regions but provide constrained spatial resolution compared to the dimensions of the observed objects, making it more challenging to construct high-precision data-driven models.

A. Motivation

HSI SR methods based on CNNs can be primarily categorized into three types: 2-D CNN, 3-D CNN, and 2-D/3-D hybrid CNN [15]. The 2-D CNN approach, similar to RGB image SR, mainly focuses on spatial features but with relatively lower spectral fidelity [16], [17], [18]. The 3-D

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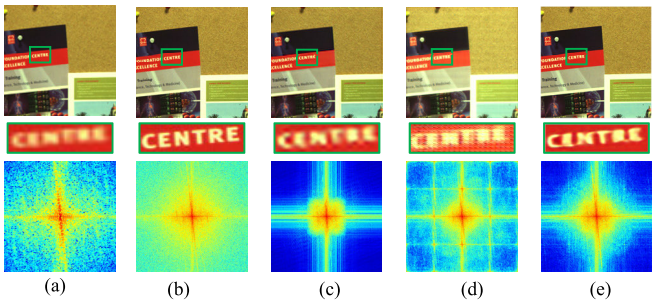


Fig. 1. Performance of several algorithms is compared on the *imgg0* image (Harvard dataset ×4). Note that the last row represents a spectrum. In the spectrogram, (c) lacks high-frequency details, while (d) introduces visible checkerboard artifacts, resulting in blurred images. In contrast, (e) uses frequency domain loss to effectively preserve high-frequency information and enhance HSI clarity. In (c), the PSNR is 38.99 dB; in (d), the PSNR is 39.18 dB; and in (e), the PSNR is 40.43 dB. (a) Input (LR) 128 × 128, (b) Ground truth 512 × 512, (c) GDRRN 512 × 512, (d) EUNet 512 × 512, and (e) SRDNet (ours) 512 × 512.

CNN method considers spectral dimensions, enhancing spectral performance but having limitations in improving spatial resolution [19], [20], [21]. In contrast, the 2-D/3-D hybrid CNN leverages both spectral information and strengthens the learning of spatial features [16], [19], [22], [23]. Inspired by these studies, this article constructs a network using a 2-D/3-D hybrid convolution. Different from many existing algorithms, considering the high dimensionality of HSI, it is difficult to capture complex mappings from low to high spatial resolution with one-step upsampling methods [24], [25]. Conversely, the progressive upsampling strategy has been proven effective in previous research [14], [26]. We aim to provide a new solution for HSI SR to further enhance its performance and effectiveness. Therefore, we adopted the progressive upsampling strategy.

Another critical issue is that existing methods typically emphasize spatial-domain considerations when addressing spatial-spectral correlations in HSI, often relying on spatial-domain loss functions. However, the frequency-domain coherence properties of HSI have not been adequately addressed. Some works [27], [28] have indicated that because of the built-in tendencies of CNNs, these approaches often prioritize modeling low-frequency elements, which are simpler to generate, possibly resulting in the omission of high-frequency features. We visualized the reconstructed HSI in Fig. 1. It is evident that, after undergoing SR algorithm processing, the texture information of the LR image does indeed improve to some extent. Nevertheless, significant frequency-domain differences persist when compared to the ground truth. For instance, EUNet [9] introduces observable checkerboard artifacts in the images, while GRNND [16] loses high-frequency information, resulting in relatively blurry reconstructed images. In contrast to these methods, our algorithm significantly improves the visibility of details in HSI by introducing frequency-domain loss constraints, effectively preserving high-frequency information. Therefore, we propose transforming HSI data into a frequency-domain representation and incorporating additional modules to compel the model to adaptively recover high-frequency information. This dual-domain strategy, which integrates both spatial and frequency domains, is expected to more accurately restore and enhance feature information in images.

Furthermore, HSIs exhibit sparsity, and features at different locations and channels have varying importance [29]. While some approaches have tried to integrate self-attention mechanisms, they often sacrifice feature detail to reduce the computational demands of non-local self-attention maps [30], [31]. These processes unavoidably degrade spectral autocorrelation and information continuity. Drawing from the widespread adoption of pixel-level attention in intricate visual applications [32], we acknowledge the importance of well-crafted HR and finely tuned spectral-spatial attention for HSIs.

B. Challenges

In HSI SR reconstruction, when employing a hybrid convolution model that combines both spatial-domain and frequency-domain strategies, the following main challenges are encountered.

1) *Data Scarcity:* Limited HSI data samples make it challenging to train complex models. We need to design more efficient network structures with the available data samples to prevent overfitting.
2) *Structural Innovation:* 2-D CNNs enhance spatial resolution while 3-D CNNs emphasize spectral information extraction. An innovative structural design that harmoniously integrates these two processes within a single model is needed to complement each other and achieve superior performance. This novel design can deliver better performance than previous studies involving 2-D/3-D hybrid CNNs.
3) *Frequency Domain Consistency Modeling:* Existing methods pay limited attention to the frequency domain consistency of HSI, which can lead to discrepancies between generated images and the ground truth. How to design a network architecture to effectively capture frequency domain information and introduce frequency domain loss constraints during the training process to enhance the performance of HSI is a challenge.
4) *Feature Adaptive Modeling:* CNN methods have constraints when it comes to capturing extended-range correlations and spectral self-similarity. Attention mechanisms are often introduced to model these features. However, enhancing the attention on certain features may reduce the resolution of others, resulting in information loss. An adaptive attention mechanism is necessary to effectively model these features without compromising their resolution.

C. Contributions

To address the aforementioned challenges, we propose a novel approach called SRDNet for HSI SR. This manuscript makes the following key contributions.

1) A hybrid convolution of 2-D and 3-D units with progressive upsampling is designed. The 2-D unit enhances

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spatial resolution, while the 3-D unit extracts spectral information.

2) To enhance the internal and external relationships between channels, an IGM with symmetric group convolutional blocks and complementary convolutional blocks is employed in the 2-D unit.
3) In order to better utilize the spectral prior to enhancing the consistency between spatial and spectral information, we introduce a dual-domain network. In the spatial domain, a self-attention mechanism (HSL) with a pyramid structure is designed to globally model the features. In the frequency domain, frequency loss (HFL) is used to optimize the model through a dynamic weighting mechanism.

The article is organized as follows: Section II provides an overview of related work on HSI SR. In Section III, we describe the proposed SRDNet method, including the network structure and the dual-domain mechanism. Experimental parameter settings and analysis are presented in Section IV. Last, Section V concludes the article.

II. Related Work

Some of the works most relevant to our approach are briefly reviewed, including CNN-based methods, attentional mechanisms, and the use of frequency domain analysis in SISR.

A. Single HSI SR

Previous studies mainly formulated HSI SR as a constrained optimization problem, where some priors were used to limit the space [33]. For instance, to explore the non-local similarity in the spatial domain, Wang et al. [34] modeled three properties of the HSI: the non-local similarity in the spatial domain, the global correlation in the spectral domain, and the smooth structure of the spatial-spectral domain [35]. Huang et al. [36] used sparse and low-rank properties to reconstruct HSI with spatial SR.

Recently, CNNs have gained more popularity than traditional optimization-based solutions for SR, due to their strong feature extraction and representation abilities. Xie et al. [37] and Yuan et al. [38] employed DCNN networks with non-negative matrix decomposition to preserve the spectral properties of the intermediate results for HSI resolution. However, DCNN networks have difficulty in fully exploiting the spatial and spectral properties of HSI, due to the large number of spectral dimensions and the lack of enough training samples. Li et al. [16] proposed a grouping strategy, called the GDDRN method with a recursive module, to better capture the correlation between spectral bands. The method combines a spectral angle mapper (SAM) with mean squared error (MSE). However, its loss function affects the enhancement of spatial resolution. Mei et al. [19] developed a neural network, named 3-D-FCNN, which used 3-D convolution to jointly explore spatial texture features and spectral correlations. Although it reduces the distortion of the spectrum, it is computationally expensive at a high upsampling factor. Hu et al. [39] integrated a collaborative non-negative matrix factorization (CNMF) strategy with the outputs of a deep feature extraction network for learning. The method works well, but it considers spatial and spectral features separately and relies too much on manual intervention. Zheng et al. [18] designed a separable spectral convolution to obtain information from each spectral band. However, this could generate numerous redundant features across hundreds of spectral bands, making it challenging to identify the most representative ones. Despite the good performance of EUNet Liu et al. [9] proposed by a small number of parameters, it is not the optimal choice. This method fails to fully utilize spectral information and instead focuses excessively on extracting spatial features, resulting in poor SAM scores. To address interspectral distortion, Li et al. [6] proposed a 3-D generative adversarial network with an attention mechanism. However, traditional 3-D convolution kernels resulted in a large number of network parameters in [6]. To overcome this issue, subsequent studies [20], [40], [41], [42]. employed separable convolution kernels, such as $1 \times k \times k$ or $k \times 1 \times 1$, instead of $k \times k \times k$. These approaches effectively reduced the model parameters and enabled the design of deeper networks. To further enhance the extraction of spatial and spectral features, Li et al. [21] proposed a hybrid network that combines 2-D convolution with 3-D separable convolution. However, it is worth noting that this approach is computationally expensive and introduces information redundancy in the network structure.

These HSI SR methods have achieved remarkable results, but they have some limitations. On one hand, the existing methods focus on learning the spatial-spectral information of HSI from the spatial domain. We address the challenge of synthesizing high frequencies by applying a separate module to the SR results. On the other hand, some algorithms do not selectively concentrate on important features and lack a global context to model spectral dependencies. To utilize the features more effectively, we devise a novel spatial-spectral self-attention mechanism in our method.

B. Attention Mechanism

The attention mechanism is a signal processing mechanism that scans the image and allocates more attention to the focus of attention. It attends to the details of the target and suppresses other irrelevant information. For instance, Woo et al. [43] introduced complementary attention along spatial and channel axes to improve network representation. Hu et al. [44] designed a SENet network with a global loss function to address spectral band correlations. These mechanisms can enhance SR by emphasizing important details and reducing noise. To further emphasize spectral characteristics, some RGB SR methods incorporate attention mechanisms from other vision tasks [45]. Dai et al. [46] introduced second-order channels to capture long-distance spatial information. Lu et al. [47] tailored an attention network for SR, considering its unique characteristics. Chen et al. [45] proposed an SR reconstruction attention mechanism based on feature maps to improve low and high-frequency feature components.

Attention mechanisms, known for their representational capabilities [6], have been applied to HSI SR. However, many

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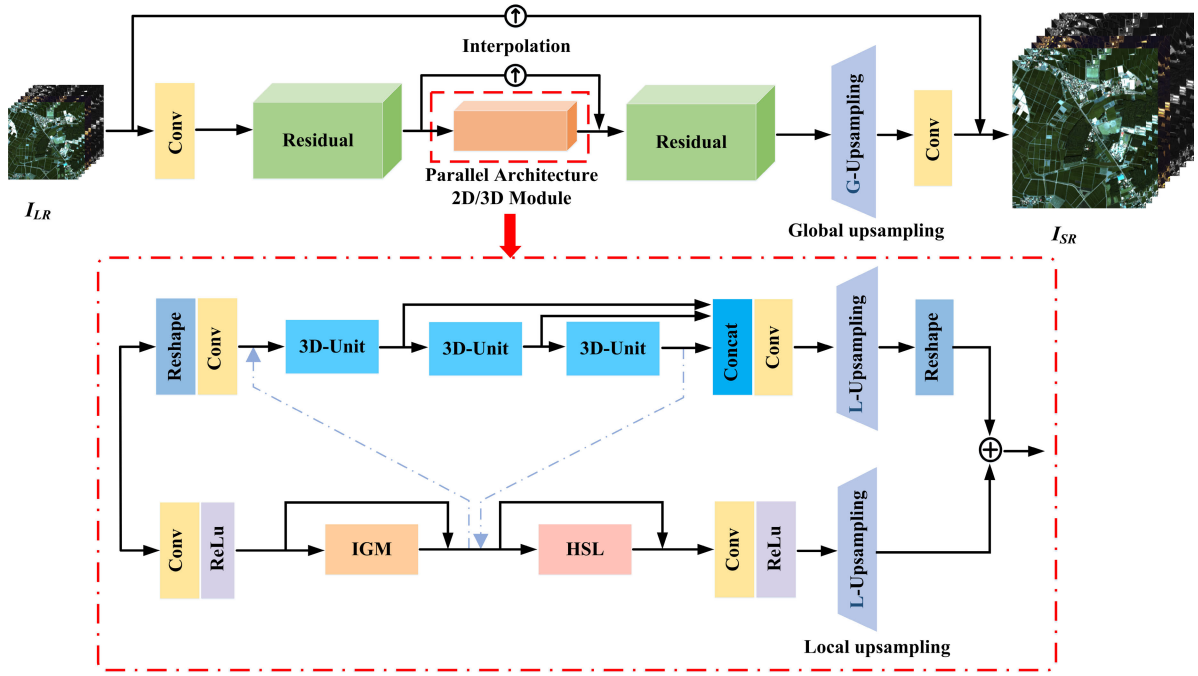


Fig. 2. Overall architecture of the SRDNet.

existing methods sacrifice internal resolution for computational speed, leading to reduced performance. Some pixel-level attention mechanisms designed for advanced tasks can further enhance model representation. For example, Liu et al. [48] devised the polarized self-attention block for pose estimation and semantic segmentation. This block preserves internal resolution during channel and spatial attention computations, achieving non-linear fitting of fine-grained regression output distributions. Fu et al. [49] introduced a dual-attention network employing self-attention mechanisms to simulate semantic interdependencies in both spatial and channel dimensions, thereby enhancing semantic segmentation performance. Xie et al. [50] proposed a clustering feature-constrained multiscale attention network for shadow extraction from remote sensing images. The method employs pixel-level reweighting mechanisms and pixel group features to establish clustering relationships between pixels, enhancing the network's capability for feature extraction. Zhang et al. [51] presented an adaptive attention mechanism, extending traditional attention mechanisms to determine the correlation between attention. Given the excellent performance of this pixel-level attention, it is imperative to explore pixel-level attention mechanisms to effectively enhance HSI SR reconstruction.

C. Image Frequency Domain Analysis

Spectrum analysis can decompose a complex signal into simpler signals. F-Principle [52] showed that deep learning methods tend to focus on low frequency to reconstruct targets, which causes differences in the frequency domain. In recent years, many CNN-based methods have been proposed to analyze the frequency domain. A coordinate-based MLP with Fourier transform [53] was used to recover high frequencies missing in single image reconstruction. Recent works have shown that frequency analysis can be integrated with SR [54], [55]. For example, some works have tried to reconstruct better visual images by minimizing the frequency domain difference between input and output during training [56]. In HSI SR, where the model is more likely to concentrate on low-frequency pixels in the spatial domain, the composite image becomes more blurred. Therefore, exploring an adaptive constraint based on intrinsic frequency is essential for reconstructing fine images.

III. Proposed Method

A. Overall Architecture

In this section, we present the detailed overall architecture of SRDNet, and show its algorithm model diagram in Fig. 2. Our method consists of four main components: the shallow feature extraction, the deep spatial-spectral feature extraction, the upsampling, and the reconstruction part. For HSI SR, we denote the LR of input and the reconstructed SR image as $I_{\mathrm{LR}} \in \mathbb{R}^{B\times H\times W}$ and $I_{\mathrm{SR}} \in \mathbb{R}^{B\times rH\times rW}$, respectively. The W and H are the width and height of the HSI, and B is the number of spectral bands in the HSI. The r is the SR scale factor that is generated by the LR. Our aim is to reconstruct the SR (I_{SR}) end-to-end by the network (SRDNet) from the LR (I_{LR}), as follows:

$$I_{\mathrm{SR}} = H_{\mathrm{Net}}(I_{\mathrm{LR}}) \tag{1}$$

where H_{Net} denotes the function of the proposed SR network. The flow of the overall network architecture is as follows.

First, the convolutional layer and the first residual block are defined as F_{conv} and $F_{\mathrm{Res}}^{(1)}$, respectively, through which the shallow features are extracted. The corresponding feature x_0 is defined as follows:

$$x_0 = F_{\mathrm{Res}}^{(1)}(F_{\mathrm{conv}}(I_{\mathrm{LR}})). \tag{2}$$

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Second, the parallel structure 2-D/3-D module (PAM) and the second residual block ($F_{\text{Res}}^{(2)}$) are adopted to extract the deep features. To alleviate the affordability of the final SR reconstruction, the strategy of progressive upsampling is applied in this article, which is divided into local upsampling and global upsampling. The $F_{L_\text{up}}(x_0 \uparrow)$ denotes the local bicubic operation. The corresponding feature x_t is defined as follows:

$$x_t = F_{\text{Res}}^{(2)}(F_{\text{PAM}}(x_0) + F_{L_\text{up}}(x_0 \uparrow, r)). \tag{3}$$

Finally, there are the upsampling and reconstruction stages, aimed at upscaling the acquired features to the target size. The upsampling module is employed to generate a spatial-spectral feature map for the target. The corresponding operation is defined as follows:

$$x_{\text{rec}} = F_{\text{conv}}(F_{G_\text{up}}(x_t, r)) \tag{4}$$

where x_{rec} is represented by convolutional layer reconstruction, F_{G_up} denotes the global upsampling operation, and the transposed 2-D convolutional layer is applied to upsample the feature map by the scale factor r

$$I_{\text{SR}} = x_{\text{rec}} + F_{\text{up}}(I_{\text{LR}} \uparrow, r) \tag{5}$$

where $F_{\text{up}}(I_{\text{LR}} \uparrow)$ represents the input (I_{LR}) for upsampling operation by bicubic interpolation. Equation (5) is equivalent to (1).

The overall network architecture has been outlined. We will delve into the sub-modules of the network, covering both the spatial-spectral and frequency domains. Specifically, we will elaborate on the parallel 2-D/3-D architecture module in the spatial domain, and provide a detailed description of the hyperspectral frequency loss (HFL) in the frequency domain.

B. Parallel Architecture 2-D/3-D Module (PAM)

After shallow feature extraction, the information flow is divided into two branches, with the first being a 3-D convolutional branching sub-network, and the second being a 2-D convolutional branching sub-network.

1) Three-Dimensional Unit Branch Network: As mentioned in Section I-A, HSIs have extra spectral dimension compared to RGB images, which allows the use of 3-D convolution to gain information beyond the spatial dimension. This article makes use of separable 3-D convolution (the convolution kernel of filter $k \times k \times k$ is replaced with $k \times 1 \times 1$ and $1 \times k \times k$) instead of the conventional 3-D convolution. To be able to use 3-D convolution, since the size of the input HSI is $B \times W \times H$, the I_{LR} is reshaped into 4-D ($C \times B \times W \times H$), and C is the number of intermediate feature channels

$$y_0^{\text{3-D}} = f_{1\times1\times1}^{\text{3-D}}(f_{Reshape}(x_0 + y_{\text{IGM}}^{\text{2-D}})) \tag{6}$$

where the $y_{\text{IGM}}^{\text{2-D}}$ repents the output of the IGM, f_{Reshape} denotes the extended dimension of the feature map extracted from the first residual group block. As for the 3-D unit, as shown in Fig. 3. The kernel is the same for each 3-D unit, and in the case of the first 3-D unit, the output can be represented as follows:

$$y_{\text{1-D}}^{\text{3-D}} = \sigma(f_{3\times1\times1}^{\text{3-D}}(\sigma(f_{1\times3\times3}^{\text{3-D}}(y_0^{\text{3-D}})))). \tag{7}$$

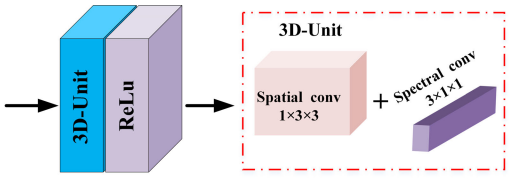


Fig. 3. Architecture of the 3-D unit.

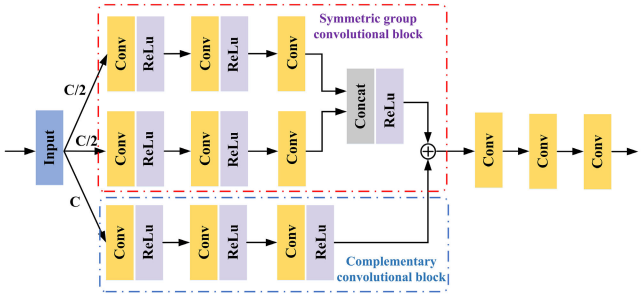


Fig. 4. IGM structure diagram.

The output of the previous 3-D unit is employed as input to the next 3-D unit. Assuming that there are N 3-D units in the model, the output is shown as follows:

$$y_{\text{ND}}^{\text{3-D}} = f_{\text{ND}}^{\text{3-D}}(f_{(N-1)D}^{\text{3-D}}(\cdots f_{\text{2-D}}^{\text{3-D}}(f_{\text{1-D}}^{\text{3-D}}(y_0^{\text{3-D}}))\cdots)) \tag{8}$$

where $f_{\text{ND}}^{\text{3-D}}$ is the operation of the Nth 3-D unit (in order to make the network simple and effective, here $N = 3$).

After obtaining hierarchical features from N 3-D units, their outputs are combined or cascaded to enable the network to capture more valuable information. Subsequently, local up-sampling is carried out, and the corresponding feature operations are as follows:

$$y_{L_\text{up}}^{\text{3-D}} = f_{L_\text{up}}^{\text{3-D}}(\sigma(f_{1\times1\times1}^{\text{3-D}}(f_{\text{Concat}}^{\text{3-D}}[y_{\text{1-D}}^{\text{3-D}}, y_{\text{2-D}}^{\text{3-D}}, y_{\text{1-D}}^{\text{3-D}}])))). \tag{9}$$

Among them, the $f_{L_\text{up}}^{\text{3-D}}$ is represented a local up-sampling operation. Finally, the obtained feature $y_{L_\text{up}}^{\text{3-D}}$ is reshaped from its 4-D representation ($C \times B \times W \times H$) to a 3-D image size ($B \times W \times H$). The output of the 3-D unit branch is obtained as follows:

$$y^{\text{3-D}} = f_{\text{Reshape}}(y_{L_\text{up}}^{\text{3-D}}) \tag{10}$$

where $y^{\text{3-D}}$ is the dimension of the feature being squeezed.

2) Two-Dimensional Unit Branch Network: In contrast to the 3-D unit operation, the 2-D unit branching network is used to extract deeper features through the IGM unit. Its structure is shown in Fig. 4.

a) Isomeric group module (IGM): Low-frequency information usually refers to smooth regions and global structures in an image that are present at different scales and orientations [57]. Shallow-level low-frequency information can often be extracted using simple convolutional layers [58]. Current HSI SR methods typically achieve resolution enhancement by grouping in the spectral dimension [14], [16] or directly fusing features across all channels [9], [21], [59], incorporating attention mechanisms for global exploration. However, this may lead to feature simplification and reduced robustness, particularly in complex HSI SR scenarios (e.g., [9],

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[16]). Simultaneously, it introduces additional computational burden, diminishing the practicality of the algorithm, such as [14], [59].

To address this issue, we introduce IGM, which boosts both the internal and external relationships among diverse channels in parallel. This enhances the network's expressive capability and captures more representative structural information. While increasing the network depth can mine more useful features, it is inspired by the GoogLeNet [60], which suggests that overly deep networks may lead to performance degradation, especially with a limited number of samples. Moreover, considering that the whole network also contains other modules, we have to strike a balance between performance and risk of overfitting. Therefore, based on the above considerations, we chose to build the IGM with fewer layers to ensure performance stability.

The IGM is composed of two components: a symmetric group convolution block and a complementary convolution block, as shown in Fig. 4. The symmetric group convolution block contains two branch structures, each with the same structure and half-channel inputs. Notably, these two branches receive input channels from the upper and lower halves of all channel features, respectively, as shown in (12)–(14). In the symmetric group convolution block, the input image is subjected to a 3×3 convolutional operation, and then a ReLU activation function is applied. This transforms the input image into nonlinear low-frequency features, facilitating the capture of smooth image features associated with low-frequency information. The outputs of these two branch networks are then concatenated to enhance SR performance. The complementary convolution block shares a similar design with the symmetric convolution block, but it differs in having a full-channel input. This enables it to process image features more deeply, including complex and global structural information. This capability is crucial for handling low-frequency features in images that exhibit variations across channels.

Finally, to mitigate the impact of redundant information introduced by the IGM and to facilitate better integration of layered features, enhancing the correlation between shallow and deep features, we introduce a module comprising three convolutional layers. The parameters for these three convolutional layers are configured as follows: input channel 64, output channel 64, and convolutional kernel size 3×3. This module conducts spatial superposition fusion operations on the outputs of the symmetric group convolution block and the complementary convolution block, harmoniously combining this information through residual connections. This empowers the IGM unit to extract information more efficiently. Next, we will delve into the specific operations and principles of symmetric convolution and complementary convolution in detail.

Symmetric Group Convolution Block: This block consists of two subnetworks with three layers each, which extract the main channel information from the image. The features obtained by the subnetworks are combined by a cascading operation to enhance their internal coherence. Each layer of the subnetworks follows the Conv+ReLU structure, utilizing 32 input and output channels along with 3×3 convolutional kernels. The symmetric group block output results from concatenating the outputs of the two subnetworks, resulting in 64 output channels

$$x_0^{\text{2-D}} = \sigma(f_{\text{conv}}^{3\times3}(x_0)) \in \mathbf{R}^{C,H,W} \tag{11}$$

$$x_0^{U} = x_0^{\text{2-D}} \in \mathbf{R}^{U,H,W} \tag{12}$$

$$x_0^{L} = x_0^{\text{2-D}} \in \mathbf{R}^{L,H,W} \tag{13}$$

$$C = U + L \tag{14}$$

where, U represents the upper half of features across all channels and L represents the lower half of features across all channels. Specifically, U and L both equal half of C, and C denotes the total number of channels. Additionally, x_0 signifies the output features after the network passes through the first residual block, as described in (2). x_0^U denotes the input features obtained by separating the upper half of channels, while x_0^L refers to the input features obtained by separating the lower half of channels.

The structures of the two subnetworks are identical. Taking the first subnetwork as an example, its output is as follows:

$$y_{1b}^{\text{2-D}} = \sigma(f_{\text{conv}}^{3\times3}(x_0^U)) \in \mathbf{R}^{C/2,H,W} \tag{15}$$

$$y_{\text{SGB}-1}^{\text{2-D}} = f_{3b}^{\text{2-D}}(f_{2b}^{\text{2-D}}(y_{1b}^{\text{2-D}})) \in \mathbf{R}^{C/2,H,W} \tag{16}$$

where the $f_b^{\text{2-D}}$ denotes the operation of convolution and ReLU function per layer, for a total of three layers. The operation of the second branch subnet is the same as that of the first subnet, but the difference is that the input feature is x_0^L, and the output feature is represented as follows:

$$y_{\text{SGB}-2}^{\text{2-D}} = f_{3b}^{\text{2-D}}(f_{2b}^{\text{2-D}}(y_{1b}^{\text{2-D}})) \in \mathbf{R}^{C/2,H,W}. \tag{17}$$

The output features of the symmetric group convolutional block are obtained by cascading the output features of the two subnetworks, which is the output feature representation of this network

$$y_{\text{SGB}}^{\text{2-D}} = f_{\text{Concat}}^{\text{2-D}}[y_{\text{SGB}-1}^{\text{2-D}}, y_{\text{SGB}-2}^{\text{2-D}}] \in \mathbf{R}^{C,H,W}. \tag{18}$$

Complementary Convolution Block: This block consists of a single three-layer subnetwork that captures the total information of all channels in the image. The complementary convolution block improves the external correlation and robustness of the algorithm, which is a valuable complement to the symmetric group convolution block. Each layer of the complementary convolution block has the same structure as the symmetric group convolution block. The difference is that the input features are $x_0^{\text{2-D}}$ and each convolutional layer has 64 inputs, which allows for more fine-grained feature extraction. The output ($y_{\text{CCB}}^{\text{2-D}}$) of the complementary convolution block is shown as follows:

$$y_{\text{CCB}}^{\text{2-D}} = f_{3b}^{\text{2-D}}(f_{2b}^{\text{2-D}}(y_{1b}^{\text{2-D}})) \in \mathbf{R}^{C,H,W}. \tag{19}$$

The output of symmetric group convolution blocks and complementary convolution blocks are spatially superimposed and integrated by a convolution fusion operation, which produces the output of the IGM. The output characteristic of this module is expressed as follows:

$$y_{\text{IGM}}^{\text{2-D}} = f_{\text{conv}}(y_{\text{CCB}}^{\text{2-D}} + y_{\text{SGB}}^{\text{2-D}}) + x_0^{\text{2-D}} \in \mathbf{R}^{C,H,W}. \tag{20}$$

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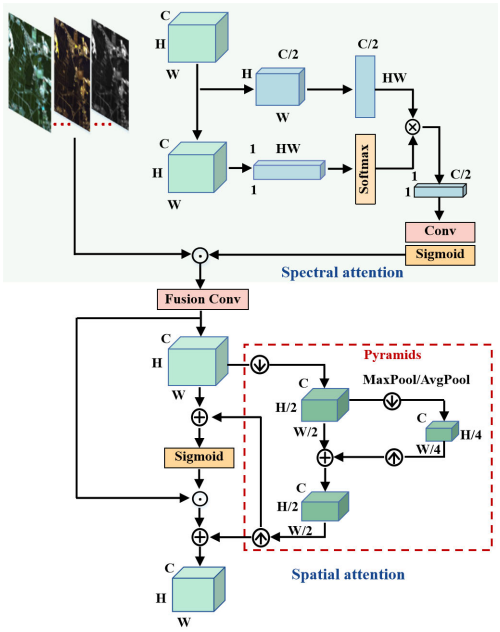


Fig. 5. Hyperspectral attention mechanism.

b) HSL module: To begin with, the indiscriminate treatment of all types of features in the convolution kernel limits the ability of the network to reconstruct. It is difficult to distinguish useful high-frequency information from rich low-frequency features [61]. Moreover, the perceptual field of each layer of the convolutional kernel is only a local expression of the fact that the output is only partially related to the input, and it lacks sufficient contextual information. Furthermore, the spectral features are of great importance for the HSI, so an adequate exploration of spectral correlations cannot be neglected. Based on the above three considerations, we incorporate attention mechanisms from the spatial and spectral dimensions, respectively. This allows the exploration of the long-range dependence of pixel positions and the correlation of spectral bands. The specific structure is illustrated in Fig. 5.

Spectral Attention: The convolutional layer is utilized to linearly map the input feature to obtain the *query* vector and *key* vector. The full space *query* vector and half channel *key* vector are obtained by reshape operation. Then, the *query* vector is remapped to the *key* vector through matrix multiplication. The auto-correlation of the spectral dimension is calculated to obtain the attention coefficient, which is recorded as the *value* vector. Its channel is maintained as C/2, which avoids excessive computational costs

$$y^{\text{2-D}}_{\text{Spectal-}q} = f_{\text{Soft max}}(f_{\text{Reshape}}(f_{\text{conv}}(y^{\text{2-D}}_{\text{IGM}} + y^{\text{3-D}}_{\text{3-D}}))) \in \mathbf{R}^{1,\text{HW}} \tag{21}$$

$$y^{\text{2-D}}_{\text{Spectal}-k} = f_{\text{Reshape}}(f_{\text{conv}}(y^{\text{2-D}}_{\text{IGM}} + y^{\text{3-D}}_{\text{3-D}})) \in \mathbf{R}^{C/2,HW} \tag{22}$$

$$y^{\text{2-D}}_{\text{Spectral}-v} = y_{\text{Spectral-}q} \otimes y^{\text{2-D}}_{\text{Spectral}-k} \in \mathbf{R}^{C/2,1,1} \tag{23}$$

where the $\otimes$ denotes the matrix multiplication operation and the f_{Softmax} represents the softmax function. As shown in Fig. 5, after the value vector is activated by convolution and sigmoid, the weight coefficient of each channel can be obtained as follows:

$$y^{\text{2-D}}_{\text{Spectral-map}} = f_{\text{Sigmoid}}(f_{\text{conv}}(y^{\text{2-D}}_{\text{Spectral}-v})) \in \mathbf{R}^{C,1,1} \tag{24}$$

where f_{Sigmoid} denotes the sigmoid activation function and $f^{\text{2-D}}_{\text{Spectral}-\text{map}}$ denotes the channel weight coefficient.

Finally, the $y^{\text{2-D}}_{\text{IGM}}$ is multiplied by the weight factor of each channel to obtain the calibrated input signal, so that the HSI focuses on the channel with the greater weight.

Spatial attention: To enhance the fused features with spatial awareness, we compute the spatial attention mask based on spectral attention. Inspired by the successful application in the field of video SR [32], a pyramid structure is employed to progressively enlarge the receptive field, capturing spatial contexts at different scales. Then we use element-wise multiplication and addition operations to combine the features with the mask, fusing the features in an adaptive and effective manner. Finally, the Sigmoid function is used to map the original attention scores to a probability distribution, allowing the network to focus more on the regions of interest.

Differing from the TSA module [32], several optimizations have been implemented. First, the limitations of the module at an upsampling factor of 3 are addressed. Second, for enhanced representational capacity, unnecessary Leaky ReLU functions are removed to optimize model performance. While Leaky ReLU assists in resolving the issue of neuron death caused by traditional ReLU in negative input situations, it is not always the optimal choice. Excessive use of Leaky ReLU can lead to problems such as gradient vanishing during model training [62]. Additionally, 3×3 convolutions are utilized for spatial feature extraction with parameter sharing, significantly reducing model parameters. Last, while still using average pooling and max pooling for downsampling, we introduce concatenation operations to dynamically fuse the generated data using weights ω_1 and ω_2. Some skip connections are also introduced, allowing the current residual block to focus more on high-frequency information while directly transmitting low-frequency features to the end. The specific operations are as shown in the following formula:

$$y^{\text{2-D}}_{\text{Fusion}} = f_{\text{conv}}(y^{\text{2-D}}_{Spectral}) \in \mathbf{R}^{C,H,W} \tag{25}$$

$$y^{\text{2-D}}_{\text{Down}} = f_{\text{conv}}[\omega_1 * y^{\text{2-D}}_{\text{max}}, \omega_2 * y^{\text{2-D}}_{\text{avg}}] \in \mathbf{R}^{C,H/2,W/2} \tag{26}$$

$$y^{\text{2-D}}_{\text{Pyramid}} = f_{\text{Up}}(f_{\text{Down}}(y^{\text{2-D}}_{\text{Fusion}})) + y_{\text{Down}} \in \mathbf{R}^{C,H/2,W/2} \tag{27}$$

$$y^{\text{2-D}}_{\text{Spectral}} = y^{\text{2-D}}_{\text{IGM}} \odot y^{\text{2-D}}_{\text{Spectral-map}} \in \mathbf{R}^{C,H,W} \tag{28}$$

where the $y^{\text{2-D}}_{\text{Fusion}}$ denotes the fusion of spectrally corrected features, the $y_{\text{max}}(\cdot)$ and $y_{\text{avg}}(\cdot)$ denote the maximum pooling and average pooling operations of downsampling, respectively. The $f_{\text{Down}}(\cdot)$ denotes the acquired post-sampling features and the $f_{\text{Up}}(\cdot)$ denotes the upsampling operation.

As shown in Fig. 5, the obtained features have undergone two downsampling and upsampling, respectively. Then, according to the sigmoid function, the weight coefficient of the spatial pixel can be obtained as follows:

$$y^{\text{2-D}}_{\text{Spatial-map}} = f_{\text{Sigmoid}}(f_{\text{Up}}(y^{\text{2-D}}_{\text{Pyramid}}) + y_{\text{Fusion}}) \in \mathbf{R}^{C,H,W} \tag{29}$$

$$y^{\text{2-D}}_{\text{Spatial}} = y_{\text{Fusion}} \odot y^{\text{2-D}}_{\text{Spatial-map}} \in \mathbf{R}^{C,H,W} \tag{30}$$

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where the $f^{2\text{-D}}_{\text{Spatial}-\text{map}}$ denotes the obtained spatial pixel weighting factor. After the calibrated input signal is obtained, the HSI focuses on the pixel area of greater weight.

The corrected feature signals in the spectral and spatial domains are fused so that the output feature maps are highly correlated in both spatial pixels and spectral dimensions

$$y^{2\text{-D}}_{\text{HSL}} = (y^{2\text{-D}}_{Spectral} + y^{2\text{-D}}_{Spatial}) + y^{2\text{-D}}_{\text{IGM}} \tag{31}$$

where the $f^{2\text{-D}}_{\text{HSL}}$ denotes the output characteristics of the HSL module. Similar to the 3-D branch network, local upsampling is used to alleviate the burden of the final SR reconstruction. This is shown in the following operations:

$$y^{2\text{-D}} = f^{2\text{-D}}_{L_\text{up}}(\sigma(f_{\text{conv}}(y^{2\text{-D}}_{\text{HSL}}))). \tag{32}$$

In the PAM network, the outputs of the 3-D unit branch network and the 2-D unit branch network are merged from spatial pixels. The definition is as follows:

$$x_t = y^{2\text{-D}} + y^{3\text{-D}} \tag{33}$$

where, x_t denotes the output of the entire PAM, resulting from the operation of the $F_{\text{PAM}}(\cdot)$ function in (3).

C. Loss Function

The L_1 loss is a common choice for SR works, as it can ensure good convergence in training. In this article, we use the L_1 loss to compute the pixel loss in the spatial domain. At the same time, the L_{HFL} is used to calculate the loss in the spectral frequency domain. The total loss (L_{total}) combined with the dual-domain network is expressed as follows:

$$L_{\text{total}} = L_1 + \beta L_{\text{HFL}} \tag{34}$$

where the β is utilized to balance the contribution of various losses and the β is set to 0.1.

L1 loss

The spatial loss is constructed from the L_1 paradigm as follows:

$$L_1(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\left\|I^n_{\text{GT}} - H_{\text{Net}}(I^n_{\text{LR}})\right\|_1 \tag{35}$$

where the I^n_{GT} and the $H_{\text{Net}}(I^n_{\text{LR}})$ denote the n_{th} ground truth and the reconstructed HSI, the N indicates the number of HSI in a training batch, and the Θ is used as the parameter set of our network.

Hyperspectral Frequency Loss

In order to process and analyze the image in the frequency domain, we apply the 2-D discrete Fourier transform (DFT) to convert the HSI from a spatial domain to a frequency domain representation, as shown in Fig. 6(b)

$$F_{(u,v)} = \sum_{x=0}^{H-1}\sum_{y=0}^{W-1} f(x,y)\cdot e^{-i2\pi(\frac{ux}{H}+\frac{vy}{W})} \tag{36}$$

where the image dimensions are $H \times W$ and the (x, y) denotes the coordinate of the image pixels in the spatial domain. The $f(x,y)$ is a pixel value. The (u,v) denotes the spatial coordinate of the spectrum and the $F(u,v)$ is the complex frequency value, which is the sum of each image pixel in the spatial domain.

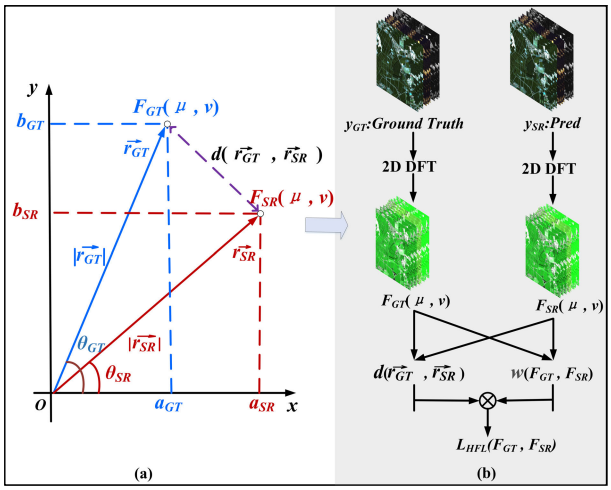


Fig. 6. Hyperspectral frequency loss. (a) Represents the loss vector map, while (b) depicts its Fourier transform.

A distance metric can be employed to assess the distinction between the reconstructed SR and the ground truth in the frequency domain, as depicted in Fig. 6(a). In the frequency domain, frequencies within the same space are manipulated and presented as distinct 2-D sinusoidal components in the image. It is essential to consider both amplitude and phase when utilizing frequency distances, as they encapsulate different features of the image. Each frequency value is mapped into a 2-D space (i.e., a plane) and transformed into a Euclidean vector. The frequency distance is then computed as the distance between the $\vec{r}_{\text{GT}}$ and the $\vec{r}_{\text{SR}}$, which involves both the angle and magnitude of the vector [the purple line in Fig. 6(a)]. We use the squared Euclidean distance for a single frequency (the kth example)

$$d^k_{(\vec{r}_{\text{GT}},\vec{r}_{\text{SR}})} = \left\|\vec{r}^k_{\text{GT}} - \vec{r}^k_{\text{SR}}\right\|^2_2 = \left|F^k_{\text{GT}}(u,v) - F^k_{\text{SR}}(u,v)\right|^2 \tag{37}$$

where $k = \{0, 1, 2, \ldots, \text{C-1}\}$, and the frequency distance between the ground truth and the reconstructed image can be written as the mean value

$$d^k_{(u,v)} = \frac{1}{HW}\sum_{u=0}^{H-1}\sum_{v=0}^{W-1}\left|F^k_{\text{GT}}(u,v) - F^k_{\text{SR}}(u,v)\right|^2. \tag{38}$$

Due to the inherent bias, the network will still be biased toward easy frequencies, so to achieve the training focus on hard frequencies, a spectral weight matrix $w(u,v)$ is introduced to weight the easy frequencies. The specific definition is as follows:

$$w^k_{(u,v)} = \left|F^k_{\text{GT}}(u,v) - F^k_{\text{SR}}(u,v)\right|^{\alpha} \tag{39}$$

where α is the scaling factor for flexibility (in the experiment α =1). Further normalize $w(u, v)$ to the range [0, 1], where 1 corresponds to the frequency that is currently lost the most and the easiest. Each frequency has different weights.

By computing the Hadamard product of the $w(u, v)$ matrix values and the frequency distance matrix [as shown in Fig. 6(b)], the output of the HFL is as follows:

$$d(F^k_{\text{GT}}, F^k_{\text{SR}}) = \frac{1}{HW}\sum_{u=0}^{H-1}\sum_{v=0}^{W-1} w^k_{(u,v)}\left|F^k_{\text{GT}}(u,v) - F^k_{\text{SR}}(u,v)\right|^2 \tag{40}$$

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$$L_{\mathrm{HFL}}(F_{\mathrm{GT}}, F_{\mathrm{SR}}) = \sum_{k=0}^{C-1} d(F_{\mathrm{GT}}^{k}, F_{\mathrm{SR}}^{k}). \tag{41}$$

IV. Experiments

A. Experiment Settings

1) Datasets:

a) CAVE database: The CAVE database[1] was collected with a cooled CCD camera [63] in the wavelength range of 400–700 nm (31 bands) in 10 nm steps. The database contains 32 object scenes with the HSI sizes of 512 × 512 × 31.

b) Harvard database: The Harvard dataset[2] was acquired using Nuance FX, CRI Inc. cameras [64] and was located in a scene of daylight or outdoor scene. The dataset contains 77 HSIs, each with a size of 1040 × 1392 × 31.

c) Chikusei database: The Chikusei dataset[3] was captured by the Headwall Hyperspec-VNIR-C sensor in Chikusei, Japan. It encompasses a wavelength range of 343 to 1018 nm and comprises 128 spectral bands. The HSIs have a spatial resolution of 2.5 m and a size of 2517 × 2335.

2) Implementation Details: Each dataset was captured using a different camera, requiring separate training and testing. In the CAVE and Harvard datasets, 80% of the datasets were randomly allocated for training, 10% for validation, and 10% for testing. During training, we randomly selected 24 patches for augmenting the training data. These patches underwent operations such as horizontal flipping (90°, 180°, 270°) and scaling (1, 0.5, 0.75). Subsequently, the patches were downsampled using bicubic interpolation with a scale factor of r, resulting in low spatial resolution images sized $32 \times 32 \times B$. The CAVE dataset comprises 8424 patches, while the Harvard dataset contains 15 160 patches. During the testing phase, we selected a 512 × 512 area in the upper left corner as the test image for evaluation.

For the Chikusei dataset, we employed the same data augmentation strategy as with the previous two datasets. We divided the images into four non-overlapping top regions, each sized at 512 × 512 × 128, which constituted the test set. In addition, overlapping patches were extracted from the remaining areas to create the training set, with 10% reserved for validation. At an upsampling factor of 4, the training set comprised 3460 patches, while at an upsampling factor of 8, it contained 834 patches. To be specific, at an upsampling factor of 4, the extracted patches were 64 × 64 pixels with a 32-pixel overlap, and at an upsampling factor of 8, they were 128 × 128 pixels with a 64-pixel overlap. Notably, the LR HSI is obtained by downsampling the patch using upsampling factors of 4 and 8, respectively.

Throughout training, the initial learning rate is set to 1.0^{-4}, and it has a total of 100 epochs. Our experiments were conducted on the Ubuntu 18.04 operating system, utilizing the PyTorch 1.7.1 deep learning framework. Hardware acceleration was performed using an RTX 3090 GPU.

[1]https://www.cs.columbia.edu/CAVE/databases/

[2]http://vision.seas.harvard.edu/hyperspec/

[3]https://naotoyokoya.Com/Download.html

3) Evaluation Metrics: To assess the performance of our network, we used four popular quantitative measures of image quality (PQI): peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) [65], mutual correlation (CC) [66], and SAM [67]. PSNR and SSIM are common metrics for image restoration quality, and their ideal values are $+\infty$ and 1, respectively. CC and SAM are frequently used in HSI fusion works, and their best values are 1 and 0, respectively.

4) Comparison With Competitive Models: In our performance comparison experiments, three categories of advanced algorithms based on CNNs are introduced, alongside the traditional Bicubic method, to highlight the exceptional performance of the SRDNet method. First, methods based on 2-D CNNs are compared, including GDRRN, EDSR, and EUNet. Second, a comparison of methods based on 3-D CNNs is conducted, which includes 3-D-FCNN. Finally, methods based on a combination of 2-D/3-D convolutions, namely MCNet, are compared. These algorithms demonstrated superior performance at various stages. With the exception of EDSR, all other algorithms are specifically designed for HSI SR. To ensure a fair comparison, experiments are conducted following the default parameters and structures recommended in the literature. Detailed information on these methods is provided below.

a) GDRRN: GDRRN adopts a global residual structure, transforming the input HSI through an embedded group recursive module, within which the kernel size of each layer is 3 × 3 × 128. This method combines traditional MSE loss with SAM loss to learn network parameters, thereby reducing numerical errors and spectral distortions in the SR results, ultimately enhancing performance.

b) EDSR: EDSR, a classic algorithm in RGB image SR, builds upon SRResNet. It employs a network with 16 residual blocks, allowing for model expansion while conserving space. This enhancement boosts expressiveness, achieving state-of-the-art performance at the time. It is widely applied in various SR domains.

c) EUNet: EUNet is a lightweight model that employs depth-wise separable convolutions to build an unfolding network for HSI SR, with an intermediate feature layer set to 128. Simultaneously, it introduces a lightweight spectral attention mechanism to capture spectral correlations. This model has demonstrated excellent performance on the Chikusei dataset.

d) 3-D-FCNN: 3-D-FCNN utilizes 3-D convolutions to construct a novel 3-D network for HSI spatial SR, comprising four 3-D convolutional layers. This approach effectively alleviates spectral distortion issues that arise when traditional CNN-based SR algorithms are directly applied to HSIs.

e) MCNet: MCNet employs four mixed convolution modules to distinguish the exploration of spatial and spectral information. By sharing spatial information, the method attempts to enhance spatial exploration while extracting spectral content simultaneously. MCNet achieves outstanding performance on the CAVE and Harvard datasets.

B. Comparisons With the State-of-the-Art Methods

Utilizing various deep learning algorithms, three datasets, namely CAVE, Harvard, and Chikusei, are utilized to validate

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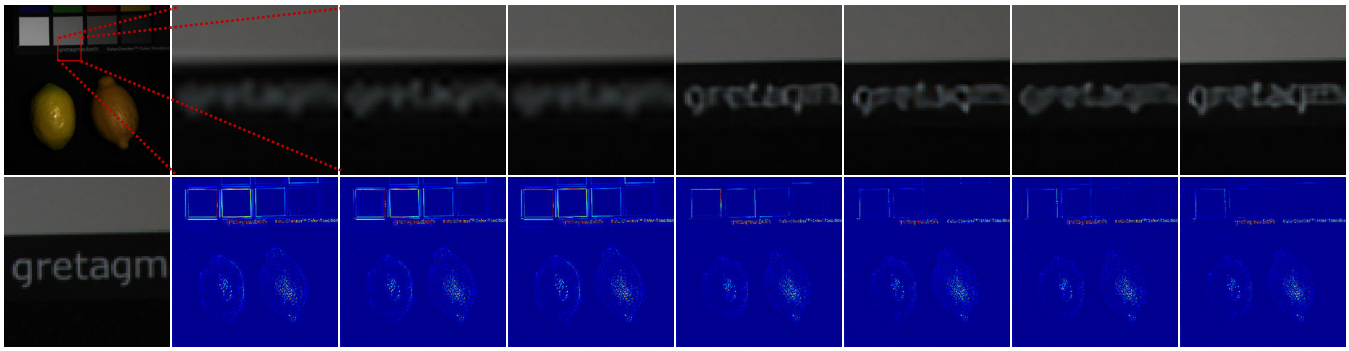


Fig. 7. First row depicts the reconstructed *fake_and_real_lemons_ms* image from the CAVE dataset, using spectral bands 23-15-8 as R-G-B, with an upsampling factor of 4. The first row is the results of different methods. The second row represents the absolute error, indicating that the reconstructed image approaches the target image more closely when it contains less information.

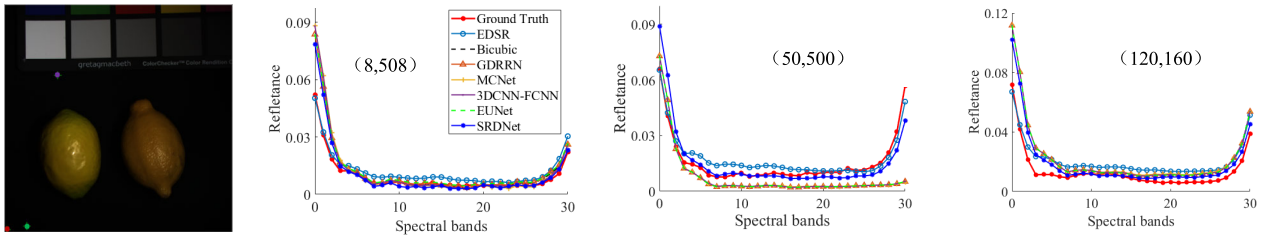


Fig. 8. Visual comparison of the *fake_and_real_lemons_ms* image spectral curves at pixel locations (8,508), (50,500), and (120,160) on the CAVE dataset.

the benefits of SRDNet across various sampling factors and bands' numbers. The specific quantitative analysis and visual evaluations are as follows:

1) Results on CAVE Dataset: As shown in Table I, our method has slightly more parameters than EDSR but fewer than MCNet. We achieved superior results under the same scaling factor. To comprehensively compare the reconstruction performance across all spectral bands, an upsampling factor of 4 is employed, and the generated HR images and spectral fitting maps are evaluated from both spatial and spectral perspectives, as depicted in Figs. 7 and 8. For improved visualization of HSI, the 23rd, 15th, and eighth spectral bands are selected as R-G-B channels to generate pseudo-color RGB images. As illustrated in Fig. 7, apart from EDSR demonstrating superior visual effects, other comparative algorithms exhibit relatively poorer performance in reconstructing fine details, as indicated by the annotated red boxes. This is attributed to EDSR's emphasis on learning spatial features with 16 residual blocks, effectively capturing deep information. However, observations from Table II and Fig. 8 reveal that due to the lack of consideration for spectral correlations, EDSR exhibits lower SAM metrics and inferior spectral curve fitting. GDRRN uses grouped convolution, which effectively reduces the network parameters, but has a relatively simple structure with only nine recursive blocks, resulting in low spatial resolution metrics and spectral fidelity. To extract features effectively, 3-D-FCNN utilizes 3-D convolutions, retaining more spectral-related information. However, its performance in spatial resolution extraction is suboptimal, leading to somewhat blurred reconstruction details. Similarly, EUNet, designed based on depthwise separable convolutions

TABLE I
COMPARISON OF ALGORITHMS IN TERMS OF NUMBER OF CAVE PARAMETERS

Methods	Scale (×2)	Scale (×3)	Scale (×4)
Bicubic	-	-	-
GDRRN[16]	0.22M	0.22M	0.22M
3D-FCNN[19]	0.039M	0.039M	0.039M
EDSR[68]	1.40M	1.59M	1.55M
MCNet[21]	1.93M	2.04M	2.17M
EUNet[9]	0.56M	0.63M	0.62M
SRDNet(Ours)	1.53M	1.61M	1.56M

(DCNNs), incorporates attention mechanisms to explore spectral correlations, showing better performance with higher upsampling factors and more spectral bands. However, its robustness is lacking, as evidenced by the SAM values in Table II, which are suboptimal for upsampling factors 3 and 4. This may be due to the use of separable convolution, resulting in relatively blurred reconstruction details, although the metrics are satisfactory. MCNet, employing a hybrid convolution approach, effectively exploits spectral correlations while reconstructing spatial resolution. However, it relies solely on spatial loss to constrain network learning, lacking frequency domain modeling, leading to suboptimal performance. In addition, its one-step upsampling strategy performs poorly with large upsampling factors.

In contrast to these algorithms, our proposed SRDNet employs both spatial and frequency domain constraints in network modeling. Simultaneously, a hybrid convolution with upsampling strategies is employed, allowing our algorithm to learn spatial features effectively while capturing spectral

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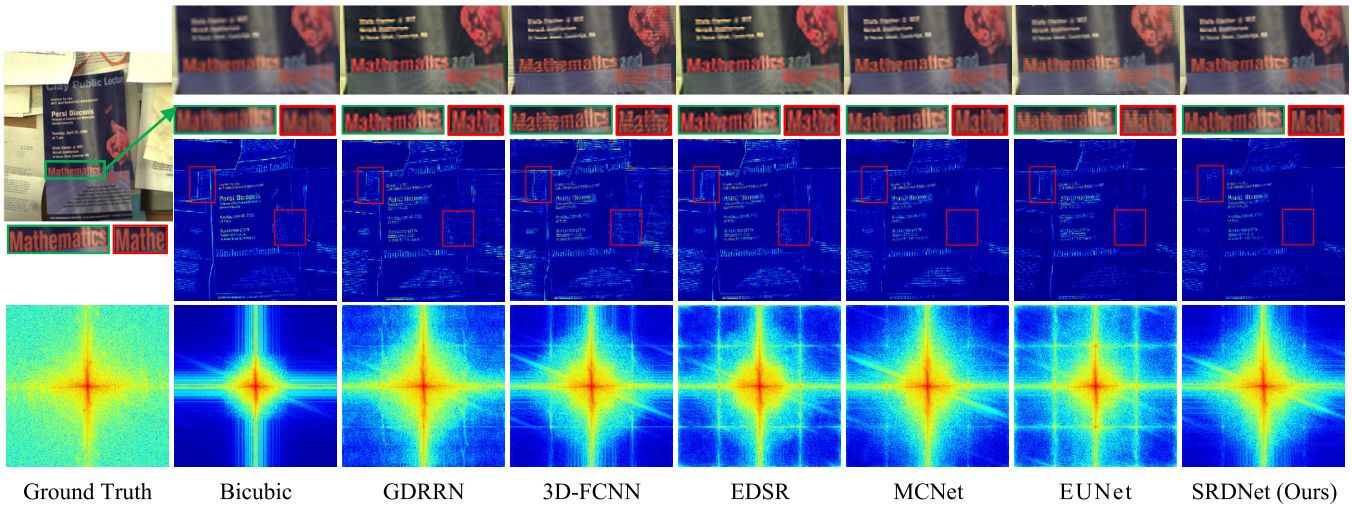


Fig. 9. First row illustrates the reconstructed *img3*image from the Harvard dataset using spectral bands 23-15-8 as R-G-B with an upsampling factor of 4. The second row displays the absolute error, demonstrating that the reconstructed image becomes closer to the target image when it contains less information. The third row represents the spectrograms.

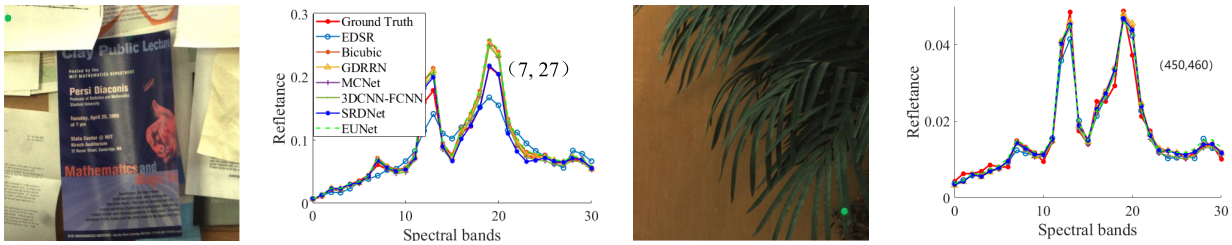


Fig. 10. This is a visual comparison of the image spectral curves on the Harvard dataset.

TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT METHODS FOR THE FOUR PQIS PAIRS OF FIVE TEST IMAGES ON THE CAVE DATASET. RED AND BLUE SHOW THE BEST AND SECOND BEST PERFORMANCE RESPECTIVELY

Methods	Scale	CC↑	SAM↓	PSNR↑	SSIM↑
Bicubic		0.990	2.605	42.662	0.9885
GDRRN[16]		0.991	2.609	43.423	0.9898
3D-FCNN[19]		0.991	2.605	43.377	0.9864
EDSR[68]	×2	0.991	2.905	44.362	0.9903
MCNet[21]		0.992	2.417	45.082	0.9915
EUNet[9]		0.992	2.541	45.261	0.9921
SRDNet(Ours)		0.993	2.214	46.424	0.9929
Bicubic		0.983	3.255	38.988	0.9763
GDRRN[16]		0.984	2.895	39.337	0.9788
3D-FCNN[19]		0.985	2.907	41.345	0.9792
EDSR[68]	×3	0.988	3.163	42.432	0.9859
MCNet[21]		0.988	2.750	42.785	0.9863
EUNet[9]		0.989	4.419	42.356	0.9862
SRDNet(Ours)		0.989	2.672	42.944	0.9869
Bicubic		0.979	3.654	36.853	0.9638
GDRRN[16]		0.980	3.622	37.164	0.9649
3D-FCNN[19]		0.981	3.421	38.471	0.9683
EDSR[68]	×4	0.985	3.258	40.135	0.9784
MCNet[21]		0.986	3.033	40.271	0.9801
EUNet[9]		0.991	5.324	40.308	0.9802
SRDNet(Ours)		0.986	3.025	40.612	0.9807

correlations. Consequently, the visual effects from Figs. 7 and 8 demonstrate that our algorithm can restore more detail information while minimizing error content. In Table II, our algorithm outperforms the second-best method in terms of PSNR and SSIM, with improvements of 1.163 dB/0.0008, 0.159 dB/0.0006, and 0.304 dB/0.0005, respectively.

2) Results on Harvard Dataset: According to the Table III, the SRDNet algorithm achieves favorable results across all metrics, similar to the CAVE dataset. The absolute error diagram and the spectral curve are utilized to highlight the disparities in edge information between the ground truth and the reconstructed HSI. Fig. 9 illustrates the spectrogram of each algorithm, where the green background represents high-frequency information and the center region indicates low-frequency information.

The bicubic method tends to lose high-frequency details during image upscaling, resulting in blurry and fuzzy enlarged images. In contrast, neural network-based SR algorithms, such as MCNet [21] and SRDNet, preserve high-frequency information better and produce sharper enlarged images. Moreover, as shown in Fig. 9, it can be observed that algorithms solely relying on 2-D convolutions, such as GDRRN and EDSR, tend to exhibit color deviations, potentially indicating insufficient utilization of spectral information. On the other hand, algorithms using only 3-D convolution, such as 3-D-FCNN, result in more blurred reconstruction details. Conversely, the combined use of 2-D and 3-D modules yields significantly better results compared to using either one alone. This improvement is attributed to the effective utilization of spectral information by 3-D convolutions to enhance feature extraction, while 2-D convolutions contribute to increased spatial resolution.

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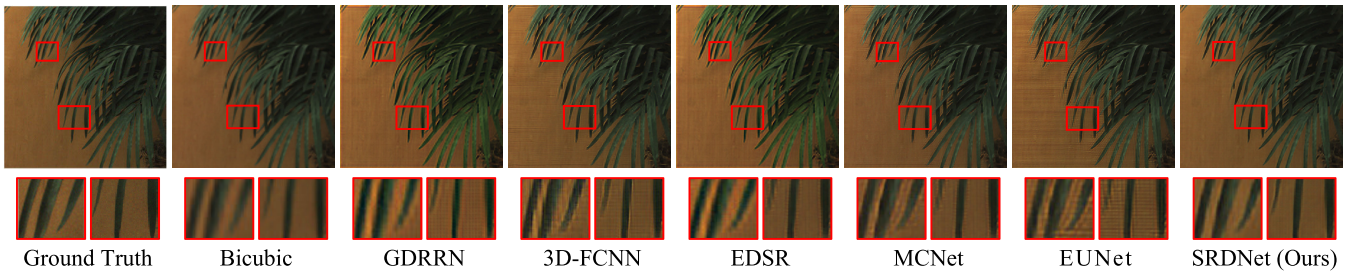


Fig. 11. Reconstructed composite images of *imgd6* in Harvard dataset with spectral bands 23-15-8 as R-G-B when the upsampling factor is 4.

TABLE III

QUANTITATIVE EVALUATION OF 25 TEST IMAGES FOR FOUR PQIS ON HARVARD DATA. RED AND BLUE SHOW THE BEST AND SECOND BEST PERFORMANCE RESPECTIVELY

Methods	Scale	CC↑	SAM↓	PSNR↑	SSIM↑
Bicubic		0.983	3.091	42.557	0.9728
GDRRN[16]		0.998	2.284	45.300	0.9889
3D-FCNN[19]		0.998	2.313	45.620	0.9874
EDSR[68]	×2	0.997	2.867	46.557	0.9808
MCNet[21]		0.998	2.236	46.845	0.9893
EUNet[9]		0.991	2.355	46.364	0.9840
SRDNet(Ours)		0.998	2.218	47.105	0.9905
Bicubic		0.979	4.262	38.714	0.9485
GDRRN[16]		0.995	3.214	41.070	0.9712
3D-FCNN[19]		0.994	3.073	40.839	0.9690
EDSR[68]	×3	0.994	3.914	40.864	0.9691
MCNet[21]		0.997	2.549	43.742	0.9837
EUNet[9]		0.986	6.295	41.519	0.9811
SRDNet(Ours)		0.997	2.443	43.951	0.9846
Bicubic		0.976	4.477	37.815	0.9397
GDRRN[16]		0.994	3.206	38.631	0.9669
3D-FCNN[19]		0.994	3.169	38.692	0.9675
EDSR[68]	×4	0.991	4.236	38.407	0.9651
MCNet[21]		0.996	2.770	42.841	0.9804
EUNet[9]		0.945	7.027	40.054	0.9756
SRDNet(Ours)		0.996	2.756	42.902	0.9813

TABLE IV

DIFFERENT APPROACHES FOR THE FOUR PQIS PAIRS OF FIVE TEST IMAGES ON THE CHIKUSEI DATASET. RED AND BLUE SHOW THE BEST AND SECOND BEST PERFORMANCE RESPECTIVELY

Methods	Scale	CC↑	SAM↓	PSNR↑	SSIM↑
Bicubic		0.883	4.317	33.869	0.8135
GDRRN[16]		0.975	3.324	36.767	0.9049
3D-FCNN[19]		0.978	3.201	37.363	0.9179
EDSR[68]	×4	0.977	3.462	37.252	0.9173
MCNet[21]		0.979	3.009	37.657	0.9196
EUNet[9]		0.981	2.762	38.050	0.9301
SRDNet(Ours)		0.981	2.659	37.872	0.9286
Bicubic		0.832	6.427	31.174	0.7438
GDRRN[16]		0.945	4.922	33.381	0.8274
3D-FCNN[19]		0.945	4.894	33.502	0.8313
EDSR[68]	×8	0.938	5.534	32.960	0.8193
MCNet[21]		0.947	4.842	33.606	0.8366
EUNet[9]		0.949	4.532	33.751	0.8422
SRDNet(Ours)		0.950	4.489	33.817	0.8455

Fig. 10 displays a comparison of spectral curves for different algorithms. Table III and Fig. 11 present the average performance of different algorithms on the Harvard dataset. SRDNet outperforms the other methods for all three upsampling factors. Particularly, our algorithm, in the case of *imgd6*, effectively preserves image edges during the reconstruction process. For upsampling factors of 2/3/4, our algorithm achieves higher PSNR and SSIM values compared to the second-best method, with improvements of 0.260 dB/0.0012, 0.209 dB/0.0009, and 0.061 dB/0.0009, respectively.

3) Results on Chikusei Dataset: The Chikusei dataset comprises 128 channels. To evaluate the performance of each algorithm under high upsampling factors, we selected upsampling factors of 4 and 8. Figs. 12 and 13 display the reconstructed images and curves. The outcomes obtained with EDSR [68], GDRRN [16], and bicubic methods are indistinct, whereas the SRDNet algorithm successfully reconstructs the primary structural features and spectral information of the image. Table IV presents the average performance of all compared algorithms on the four test images. It is evident that SRDNet outperforms all other algorithms, except for EUNet [9] when the upsampling factor is 4.

However, as discussed earlier with the previous datasets, the EUNet [9] algorithm performs poorly when dealing with a large number of images. Furthermore, the computationally intensive operations in the network architecture of the MCNet [21] can significantly prolong the convergence time of the model. The limited number of images in the Chikusei dataset also impacts the performance of algorithms. Achieving models with better generalization requires a larger number of samples for training. Although EDSR [68] networks demonstrate good performance in RGB image SR, its SAM index is relatively poor compared to single hyperspectral algorithms.

C. Ablation Study

As explained in Section III-A, our method comprises four main components, with the PAM block serving as the primary element in the feature extraction network. We refer to the network without the PAM module as the baseline. In this section, we study the effect of different combinations of PAM on model performance. Table V presents the performance of different convolution types. Overall, it can be observed from the table that traditional 3-D convolution performs slightly better than separable convolution, but its parameter and FLOPs are higher, leading to a significant increase in computational costs. For instance, at an up-sampling factor of 3, the parameter count and FLOPs of traditional 3-D convolution increase by 43% and 37%, respectively, compared to separable 3-D convolution. This excessive computational complexity may limit the deployment of the algorithm in mobile and embedded

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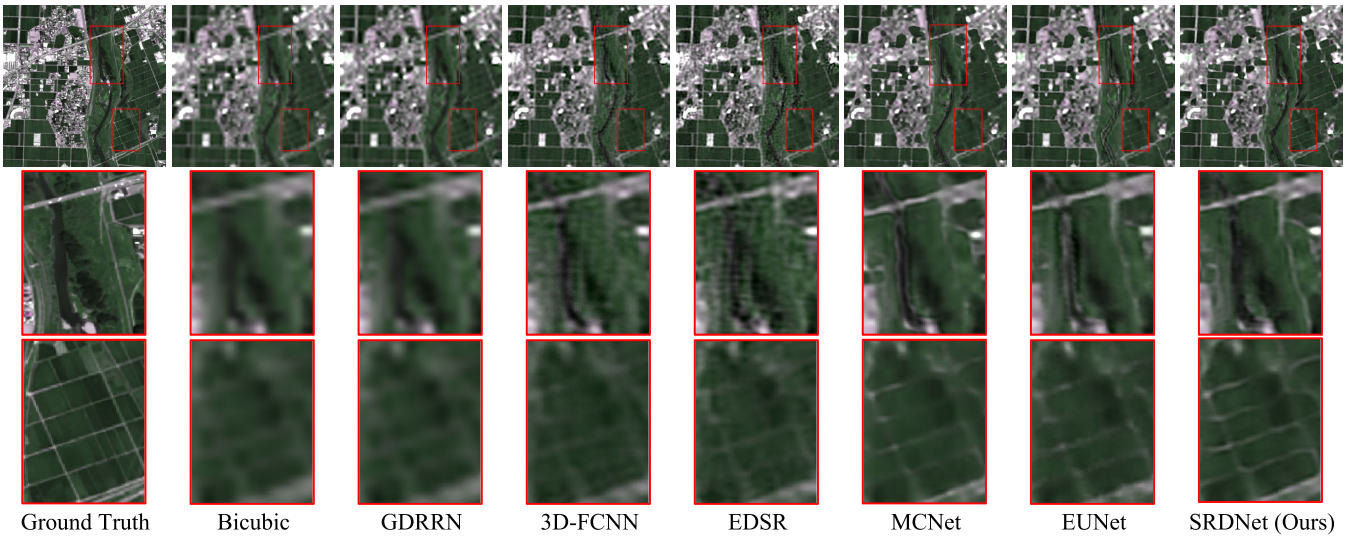


Fig. 12. Reconstructed the *Chikusei_test_2* image in the Chikusei dataset with spectral bands 55-45-55 as R-G-B when the upsampling factor is 8.

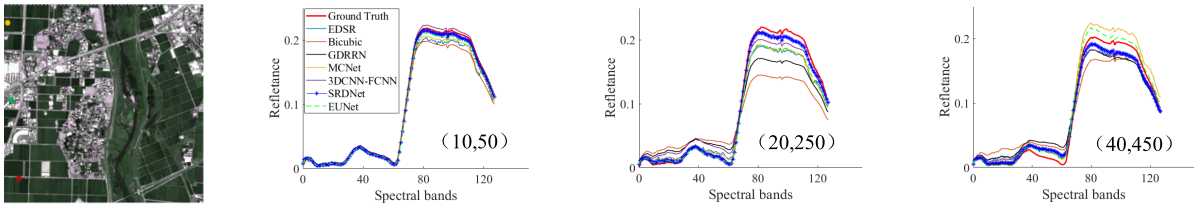


Fig. 13. Visual comparison of the *Chikusei_test_2* image (×8) spectral curves at pixel locations (10, 50), (20,250), and (40, 450) on the *Chikusei* dataset.

TABLE V

IMPACT OF DIFFERENT TYPES OF 3-D CONVOLUTION ON PERFORMANCE OF FIVE TEST IMAGES ON THE CAVE DATASET

Type	Scale	#params	#FLOPs	SAM↓	PSNR↑
Traditional3D convolution	×2	2.16M	4.32T	2.197	46.501
Separable 3D convolution	×2	1.53M	3.35T	2.214	46.424
Traditional3D convolution	×3	2.31M	4.49T	2.653	43.031
Separable 3D convolution	×3	1.61M	3.77T	2.672	42.944

TABLE VI

ABLATION STUDIES ON COMPONENTS. DIFFERENT METHODS OF FIVE TEST IMAGES ON THE CAVE DATASET

Module	Component	CC↑	SAM↓	PSNR↑	SSIM↑
3D uint(×) 2D uint(×)	Baseline	0.983	2.884	42.469	0.9847
3D uint(√) 2D uint(√)	Two 3D uint	0.987	2.741	42.925	0.9863
	Three 3D uint	0.988	2.672	42.944	0.9869
	Four 3D uint	0.987	2.679	42.937	0.9867
3D uint(√) 2D uint(×)	Three 3D uint	0.985	2.793	42.611	0.9848
3D uint(×) 2D uint(√)	IGM(√) HSL(×)	0.986	2.776	42.739	0.9859
	IGM(×) HSL(√)	0.984	2.883	42.574	0.9850
	IGM(√) HSL(√)	0.986	2.765	42.742	0.9861
HFL	L_1(√) L_{HFL}(×)	0.988	2.681	42.913	0.9869
	L_1(√) L_{HFL}(√)	0.988	2.672	42.944	0.9869

applications, such as satellites and airborne platforms. Therefore, to strike a balance between performance and lightweight design, the separable convolution is chosen. Table VI shows the ablation studies for these combinations at the upsampling factor of 3 on the CAVE dataset. Specifically, when we remove the PAM module from the network, the performance drops for the same number of parameters. When the PAM module has only 3-D units, the results are not very satisfactory. This is because the network pays too much attention to spectral information and loses spatial resolution. When the PAM has both 2-D and 3-D units, the performance is much better than only 2-D units. Additionally, in our method, a certain degree of overfitting was observed when designing four 3-D units. This could potentially be due to the limited number of images.

Fig. 14 shows that the absolute error of using the L_{HFL} loss is lower than that of using only the L_1 loss. Even compared to the classic non-local attention mechanism (Non_Local), our proposed HSL has lower absolute error information than the Non_Local [69]. To better assess the HFL module we designed, we employ the fast Fourier transform (FFT) to calculate power values at each frequency point as demonstrated as follows:

$$P_x = 10 * \log^{10}(|f_x|^2) \tag{42}$$

where the P_x denotes power values at each frequency, f_x is the result of the Fourier Transform shifted to zero frequency. The 10∗log10 operation represents taking the dB value of the energy spectrum, accentuating the differences between high and low frequencies.

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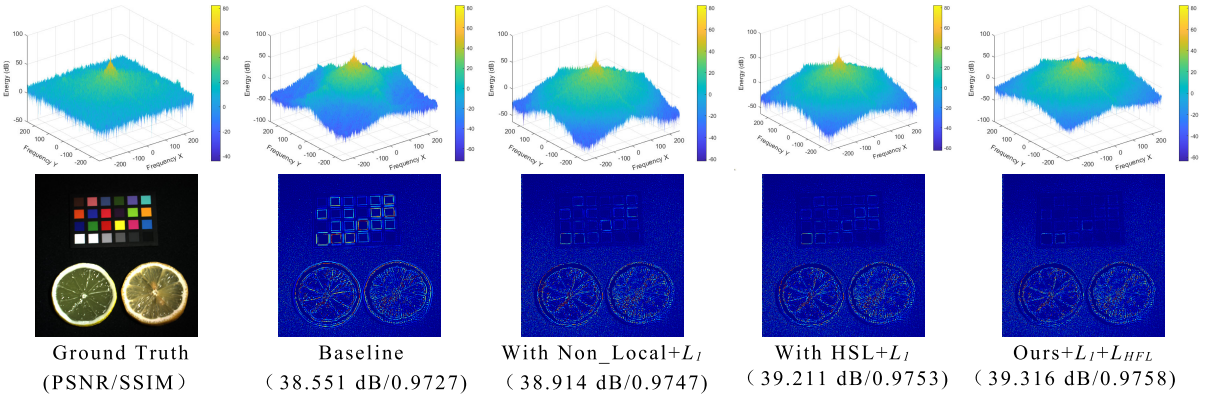


Fig. 14. First row illustrates the spectrum power comparison of the *fake_and_real_lemon_slices_ms*in the CAVE when an upsampling factor is 3. In the graph, the data from the fifth band has been selected for processing. The spectral power of this band's data has then been calculated using FFT and is displayed in units of dB (decibels). The second row displays the absolute error.

TABLE VII

ABLATION STUDY OF IGM. QUANTITATIVE COMPARISON OF DIFFERENT METHODS OF FOUR TEST IMAGES ON THE CHIKUSEI

Module	Component	SAM↓	PSNR↑	SSIM↑
IGM	SGB(×) CCB(×)	4.535	33.691	0.8420
	SGB(√) CCB(√)	4.489	33.817	0.8455
	SGB(√) CCB(×)	4.496	33.774	0.8447
	SGB(×) CCB(√)	4.499	33.763	0.8441
SGB	One branch (C)	4.493	33.789	0.8448
	two branches (C)	4.523	33.751	0.8437

SGB: Symmetric Group Convolution Block. CCB: Complementary Convolution Block. C:Number of channels.

TABLE VIII

ABLATION STUDY OF HSL,WHERE "W/O" INDICATES THAT THIS MODULE IS NOT INCLUDED, "WITH" INDICATES THAT THIS MODULE IS INCLUDED. QUANTITATIVE COMPARISON OF DIFFERENT METHODS OF FOUR TEST IMAGES ON THE CHIKUSEI

Module	Component	SAM↓	PSNR↑	SSIM↑
2D uint	w/o HSL	4.531	33.772	0.8446
	with HSL	4.489	33.817	0.8455
	w/o HSL, with Non_Local [71]	4.492	33.801	0.8451
HSL	Only spatial attention	4.493	33.806	0.8453
	Only spectral attention	4.514	33.779	0.8447

IGM Analysis

The IGM is composed of a symmetric group convolution block and a complementary convolution block, which improve channel relationships and obtain more structural information. Experiments were conducted on the Chikusei dataset (×8) to validate the effectiveness of the IGM. The results were analyzed from two perspectives. First, we removed different combinations of symmetric group convolution block and complementary convolution block separately, and the results are shown in Table VII. Second, we analyzed the impact of input channel numbers and branch network quantities in the symmetric group block while keeping the complementary convolution block unchanged. When changing the channels in symmetric convolution blocks to fully channel input ("two branches (C)") under a larger epoch setting, there was a slight performance decrease. In addition, feature extraction experiments were conducted on IGM, as shown in Fig. 15. IGM was capable of extracting certain semantic features. Importantly, the smaller the SAM value, the greater the similarity between the reconstructed HSI and the ground truth. It was observed that the SAM metric was lowest when the IGM included both. Therefore, our designed IGM structure exhibits a certain advantage.

HSL Analysis

The HSL is a self-attention mechanism that models contextual information for features globally, considering the contributions of different features to the network. It includes both spatial and spectral attention, which aids in comprehensively capturing spectral and spatial features. We compared the performance of HSL with Non_Local [69] and analyzed the individual effects of spatial and spectral attention units within HSL, on the Chikusei dataset (×8), as shown in Table VIII. The results indicate that HSL excels in various aspects. While there is only a slight improvement in PSNR compared to the "w/o HSL" approach, SAM exhibits significant enhancements. Additionally, we visualized the attention mechanism's weights on the *Chikusei_test_2* (×8) image as shown in Fig. 16. The heatmaps illustrate that our designed HSL can effectively focus on crucial features in the input data, selectively activate attention in specific regions, and make different contributions from different spectral channels through weighting.

Another aspect, for a fair comparison of our improvements, we keep our designed spectral attention module unchanged and only replace the spatial attention module with TSA [32] and our enhanced module, as shown in Fig. 17 and Table IX. From Fig. 17, it is observed that due to the excessive use of Leaky ReLU functions in TSA's spatial attention module, gradients may disappear during backward propagation, resulting in a slower descent of the training loss. In contrast, our spatial attention module removes some Leaky ReLU functions, introduces dynamic fusion weights for downsampled data, and incorporates local and global residual connections. This allows our model to enhance performance while reducing parameters, as indicated in Table IX. In terms of the training loss curve, our module exhibits a better descent rate. Due to the learning rate halving every 35 epochs in our network, fluctuations can be observed at multiples of 35 epochs. However, as the network

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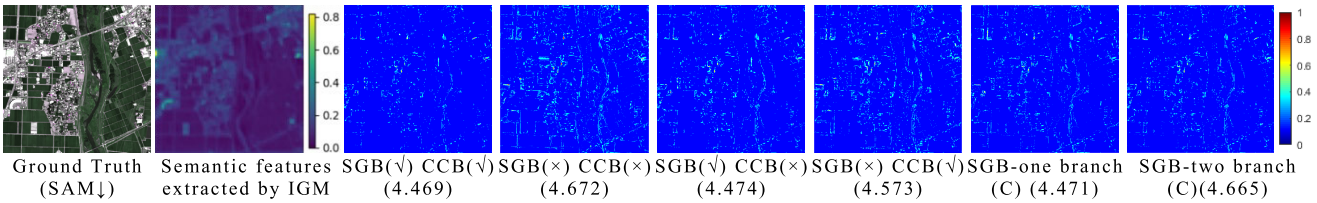


Fig. 15. Second figure shows IGM-extracted semantic features. The last six images depict SAM visualizations for different components with spectral bands 55-45-55 as R-G-B, using the *Chikusei_test_2* image (×8) in the Chikusei dataset.

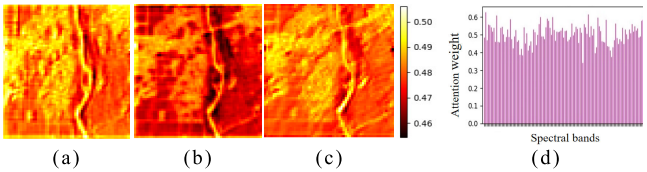


Fig. 16. Comparison of attention heatmaps on *Chikusei_test_2* image (×8). (a) and (b) Display the attention heatmaps of HSL and Non_Local [69], respectively. (c) Represents the weight heatmap of HSL with spatial attention only and (d) is the channel weights of HSL with spectral attention only.

TABLE IX
QUANTITATIVE COMPARISON OF SPATIAL ATTENTION BETWEEN TSA AND HSL IS PERFORMED ON FOUR TEST IMAGES FROM THE CHIKUSEI DATASET

Module	Component	#params	SAM↓	PSNR↑	SSIM↑
Spatial attention	TSA [32]	1.96M	4.491	33.804	0.8452
	Ours	1.79M	4.489	33.817	0.8455

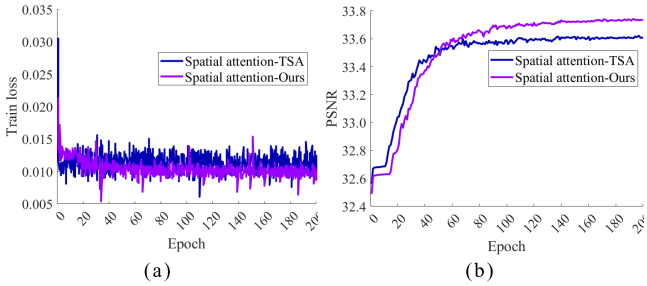


Fig. 17. Comparison is conducted on the Chikusei dataset with an up-sampling factor of 8, examining spatial attention in TSA [32] and HSL. Where (a) presents the curve of training loss and (b) shows the PSNR curves on the validation set.

gradually converges to a local optimum, the parameters stabilize. Although TSA's metrics are slightly inferior, its training is relatively more stable, attributed to the presence of more Leaky ReLU functions.

Progressive Upsampling Analysis

Inspired [26], a progressive upsampling SR framework is adopted. This approach decomposes a complex task into several simpler tasks, significantly reducing the learning difficulty while improving performance. To reduce computational complexity, a twofold upsampling was only applied in the frontend sub-network, with the remaining upsampling carried out in the final network stage ("Our-with FPU"). In Table X, we compared this method with the one that does not employ the progressive upsampling strategy ("Our-with OU"), on the Chikusei dataset. Additionally, we compared our upsampling strategy with the approach that first applies a fourfold

TABLE X
ABLATION STUDY OF PROGRESSIVE UPSAMPLING STRATEGIES, WHERE "WITH" INDICATES THAT THIS MODULE IS INCLUDED. QUANTITATIVE COMPARISON OF DIFFERENT METHODS OF FOUR TEST IMAGES ON THE CHIKUSEI DATASET

Module	Scale	SAM↓	PSNR↑	SSIM↑
Our - with PU	×4	2.659	37.872	0.9286
Our - with OU		2.919	37.688	0.9221
Our - with FPU	×8	4.489	33.817	0.8455
Our - with BPU		4.523	33.709	0.8439
Our - with OU		4.579	33.653	0.8434

PU: Progressive Upsampling. FPU: Front-end Progressive Upsampling. BPU: Back-end Progressive Upsampling. OU: One-step Upsampling.

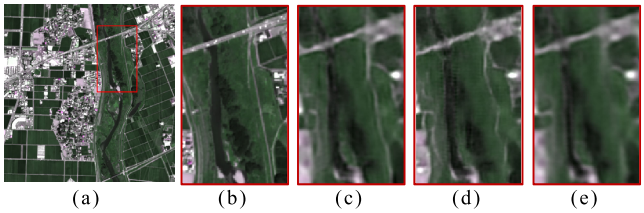


Fig. 18. Comparison of upsampling-generated reconstructed HSI on the *Chikusei_test_2* image (×8). (a) and (b) Ground truth, (c) front-end upsampling strategy, (d) back-end upsampling strategy, and (e) one-step upsampling strategy.

upsampling and then performs a twofold upsampling in the final network stage ("Our-with BPU"). The results indicate that our upsampling strategy outperforms in all metrics.

This is particularly crucial when the upsampling factor is larger, as demonstrated in Figs. 12 and 18. This might be because the progressive upsampling better captures the mapping relationship from LR to HR spatial domains [26]. It is worth noting that better visual effects can be obtained when using high upsampling ratios prematurely ("Our-with BPU") in the frontend, as shown in Fig. 18(d). However, it can lead to network overfitting in the case of limited data samples, as shown in Table X.

Finally, when we combine all components into the model, we can see that it performs better than any other combination in all three aspects. Thus, these analyses show that each component of SRDNet helps the model learn and optimize.

D. Further Discussions and Plans

Real-world HSI often faces degradation due to sensor malfunctions and atmospheric interference. For instance, the Chikusei dataset, compared to CAVE and Harvard, is heavily affected by noise, challenging the differentiation of ground objects. This leads to suboptimal reconstruction results,

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as shown in Fig. 12. Additionally, employing a larger magnification factor during downsampling exacerbates the loss of details, resulting in even worse reconstruction outcomes, as seen in Fig. 7. Our algorithm outperforms others in recovering detailed information across the three datasets. This improvement is attributed to our algorithm's use of progressive upsampling with hybrid convolutions to address severely degraded bands. The spectral attention mechanism and dual-domain loss further constrain features unfavorable for reconstruction. However, further enhancements are needed to boost our algorithm's performance, focusing on three aspects.

First, although separable 3-D convolutions excel in reducing computational costs (as shown in Table V), not all convolution kernels can be divided into two separate dimensional convolution kernels. Complete substitution of spatially separable convolutions for traditional convolutions would require searching for all possible kernels during training, potentially leading to suboptimal training solutions. Therefore, a module needs to be designed to balance the advantages of hybrid convolutions and address the limitations of separable 3-D convolutions.

Second, combining 2-D/3-D convolutions enhances network performance. Designing deeper networks becomes challenging when incorporating a 3-D structure for simultaneous input across the entire spectral range under limited hardware conditions, restricting the extraction of global features. Based on the high correlation between neighboring spectral bands, we intend to exploit the high correlation among adjacent spectral bands in future work. It aims to enhance network performance while reducing computational complexity, thereby making the network more lightweight and suitable for mobile or embedded platforms.

Finally, our PAM structure employs a parallel design, potentially introducing redundancy. To further enhance performance, we will explore the implementation of a serial structure.

V. Conclusion

A hyperspectral image SR algorithm, denoted as SRDNet, is proposed in this article. Our approach employs a hybrid convolution of 2-D and 3-D units with progressive upsampling to capture enhanced spatial information while learning spectral features. Unlike prior methods, we introduce a dual-domain learning network. The spectral self-attention mechanism is utilized to adaptively select crucial spectral information, enabling the network to acquire fine and smooth pixel-level features. Additionally, to address the visual discrepancies caused by pixel-level loss, a frequency loss is incorporated to reduce the frequency domain differences between the reconstructed image and the ground truth. The efficacy of various modules in enhancing spatial information and spectral coherence is verified through ablation studies. Visual analysis and quantitative experiments on three common hyperspectral datasets demonstrate that SRDNet excels in HSI reconstruction at both pixel and frequency levels. Our algorithm achieves optimal results on several common objective metrics, and the images are perceptually closer to the ground truth than other methods.

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